The Water Cycle

Student sheet

Name: _____ **Date:** _____

Safety: The teacher pours the warm water. The box is not waterproof: keep water away from it.

1. What I see on the screen

The screen shows **Temperature** (°C), **Humidity** (%) and **Dew point** (°C), with the chart below. You start with **Start**, move on with the green button, go back with **Back**; the last one reads **Finish**. **Humidity rises fast (20–40 s) and falls slowly (3–5 min). It is not broken. The sensor sits 5–10 cm above the water, *beside* the white steam, never in it:** droplets block it at 100% for the rest of the lesson. Do not blow towards the box.

2. What I do on my own

Water: _____ °C **Weather outside:** _____ **Windows:** ☐ open ☐ closed

First I predict, on the sheet and in the app. I do not change it after I measure.

1. Above the warm water, the humidity will reach _____ %. **2.** It will come down ☐ faster ☐ the same ☐ slower than it went up. **3.** If we warm the air **without adding any water**, the humidity will ☐ rise ☐ fall ☐ stay the same, because _____

The four zones above the vessel, read **from the bottom up**.

4	**COLD SURFACE** (the mirror)	Turns to water: **condensation**	visible? ☐ YES ☐ NO
3	**WHITE STEAM** above	**Droplets** of water, already liquid	visible? ☐ YES ☐ NO
2	**CLEAR GAP** beside the water	**Vapour** — a gas, like air	visible? ☐ YES ☐ NO
1	**LIQUID WATER** in the vessel	It all starts here: **evaporation**	visible? ☐ YES ☐ NO

The zone where **nothing is visible** is zone ___, and that is exactly where the vapour is.

How much it rises. You may tick **only the arrow**: ↑ rising, ↓ falling, → steady.

Moment	**Time**	**RH (%)**	T **(°C)**	**Arrow**
Reference, on the bench	0:00			→
Above the warm water	0:30			☐ ↑ ☐ ↓ ☐ →
Above the warm water	1:00			☐ ↑ ☐ ↓ ☐ →
Highest	_____			☐ ↑ ☐ ↓ ☐ →

How much it falls. I moved the sensor back **to the reference spot** (not by a window). I start timing again.

Moment	Time	RH (%)	T (°C)	Arrow
After the move	+1 min			☐ ↑ ☐ ↓ ☐ →
After the move	+3 min			☐ ↑ ☐ ↓ ☐ →
After the move	+5 min			☐ ↑ ☐ ↓ ☐ →

The rise. Example: $71 - 44 = 27$ *percentage points* (not percent!).
Mine: _____ − _____ = _____ percentage points.
To the peak: _____ s To return: _____ min How many times slower: _____

The two witnesses. Without them, the measurement above does not prove anything.

Witness	At the start	At the end	Difference (perc. pts)
Above the **cold** water			
By the radiator, **no water**			

By the radiator, T ☐ rose ☐ fell, and RH ☐ rose ☐ fell. **Who took the water?** _____

The dew point: the app ____ °C the spoon fogged at ____ °C table (lesson 3): ____ °C
How sure are we? Boxes side by side: ± ____ %. Warm water, twice: ____ % and ____ %

3. The conclusion I write

I predicted that, and the data show that _____

_____

Relative humidity does not say how much water is in the air, but rather _____

_____

What you see above the vessel is not vapour, but rather _____

_____

Exit ticket

One thing that surprised you today, and one thing you would do differently if you repeated the measurement.

_____